

MARÍA G. MENDOZA

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EDUCATION

- 2023–present **University of California, Berkeley** · Berkeley, CA
Ph.D., Mechanical Engineering GPA: 3.9/4.0
Advisor: Prof. S. Shankar Sastry · Research area: multi-agent systems design and control
- 2021 **Purdue University** · West Lafayette, IN
M.S., Aeronautics and Astronautics
- 2017 **University of Southern California** · Los Angeles, CA
B.S., Mechanical Engineering

RESEARCH INTERESTS

Multi-agent systems, reinforcement learning, distributed optimization, game theory, mechanism design, learning-enabled control, sequential decision-making under uncertainty, human–autonomy interaction, cyber-physical systems, and humanitarian- and societal-scale autonomy.

PROFESSIONAL EXPERIENCE

- 2017–2022 **Research Scientist** · Lockheed Martin
- ▶ Led AI/ML and robotics research projects in hypersonic systems, human–drone communication, and vehicle coordination in constrained environments.
 - ▶ Managed multi-university research collaborations (5 partner institutions) on hypersonic technologies; proposed novel aircraft-weight prediction methods.
 - ▶ Performed aerothermal analysis to develop next-generation high-speed-vehicle avionics and structures.

PUBLICATIONS

* indicates equal contribution. Full list at [Google Scholar](#).

Journal Articles

- [J1] C. Maheshwari*, **M. G. Mendoza***, V. M. Tuck*, P.-Y. Su, V. L. Qin, S. A. Seshia, ... & S. Sastry. "[Privacy-Preserving Mechanisms for Coordinating Airspace Usage in Advanced Air Mobility \(AAM\)](#)." **ACM Journal of Autonomous Transportation Systems**, 2025.

Conference Papers

- [C2] **M. G. Mendoza***, P.-Y. Su*, B. L. Ferguson, S. S. Sastry. "Dynamic Multi-Robot Task Allocation under Uncertainty and Communication Constraints: A Game-Theoretic Approach." *Under review*, 2026. Pre-print: arxiv.org/abs/2604.11954.
- [C1] **M. G. Mendoza**, A. Kalanther, D. Bostwick, E. Stephan, C. Maheshwari, S. Sastry. "[Coordinated Autonomous Drones for Human-Centered Fire Evacuation in Partially Observable Urban Environments](#)." **IEEE Global Humanitarian Technology Conference (GHTC)**, 2025.
- [C0] J. Choi, J. Aloor, J. Li, **M. G. Mendoza**, C. Tomlin, H. Balakrishnan. "[Resolving Conflicting Constraints in Multi-Agent Reinforcement Learning with Layered Safety](#)." **Robotics: Science and Systems (RSS)**, 2025.

Workshop Papers

- [W2] **M. G. Mendoza**, L. Waldburger, J. Lee, S. Sastry. [“Hierarchical Generative Agents for Simulating Sequential Human Behavior.”](#) ICLR Workshop on Multi-Agent Learning in the Era of Generative AI, 2026 (Poster).
- [W1] **M. G. Mendoza**, V. Tuck, C. Maheshwari, S. Sastry. [“Decentralized Ergodic Coverage Control in Unknown Time-Varying Environments.”](#) AAMAS Autonomous Robots and Multirobot Systems (ARMS) Workshop, 2026 (Oral).

Earlier and Other

- ▶ Multiple Lockheed Martin internal publications (classified), 2017–2022.
- ▶ “Reduction of Semi-Truck Aerodynamic Drag.” AIAA Region VI Student Paper Conference, 2017. Third-place award.

RESEARCH EXPERIENCE

- 2023–present **Graduate Student Researcher** · Sastry Lab, UC Berkeley
- ▶ Develop algorithms for modeling and controlling multi-agent systems using distributed control, optimization, learning-based methods, game theory, and mechanism design in uncertain, dynamic, and complex environments.
 - ▶ Applications: Advanced Air Mobility (AAM), autonomous robotics for humanitarian assistance, strategic vehicle path planning and dynamic routing, and safety-aware multi-agent reinforcement learning for collision avoidance.
- 2017 **Undergraduate Researcher** · IMPACT Lab, USC (Prof. Yan Jin)
- ▶ Developed and evaluated machine-learning prediction algorithms for mechanical design, including ship collision-avoidance algorithms.
- 2016 **Visiting Undergraduate Researcher** · Tsinghua University, Beijing (Prof. Mingguo Zhao)
- ▶ Designed and tested mechanical components for passive walking in bipedal robots; studied human locomotion and passive-dynamic mechanisms.
- 2014 **Undergraduate Researcher** · Materials Lab, CSU Northridge (Drs. Reiner & Bavarian)
- ▶ Battery material selection, corrosion analysis, and potentiostatic testing.

TEACHING

- Spring 2026 **Graduate Student Instructor** · UC Berkeley
EE 290 — Learning Enabled Multi-Agent Systems
- 2022–2023 **Engineering Faculty** · Cañada College, San Mateo Community College District
- ▶ ENGR -100 Introduction to Engineering
 - ▶ ENGR - 215 Computational Methods for Engineers and Scientists

INVITED TALKS AND PRESENTATIONS

- Spring 2026 **EE290 – Learning Enabled Multi-Agent Systems (Guest Lecture)**
“RESCUE-Resilient Exploration and Search Coordination of UAVs in Unknown Environments: A framework for multi-UAV coordination to assist humans and emergency responders during disasters”
- 2025–2026 **DARPA Assured Neuro-Symbolic Learning and Reasoning (ANSR) Program Meetings**
“Coordinated Autonomous Drones for Human-Centered Fire Evacuation in Partially Observable Urban Environments.”
- Oct 2025 **IEEE Global Humanitarian Technology Conference (GHTC)**
“Coordinated Autonomous Drones for Human-Centered Fire Evacuation in Partially Observable Urban Environments.”

Sep 2024	Encuentro de Mujeres Latinas en STEM del Área de la Bahía · Invited Speaker Symposium hosted by the Government of Chile. Talk: <i>“Un renacer científico en el extranjero: Una travesía entre lo profesional, la readaptación cultural y el proceso de autoconocimiento.”</i>
2017	AIAA Region VI Student Paper Conference Technical paper presentation: <i>“Reduction of Semi-Truck Aerodynamic Drag.”</i>

HONORS AND AWARDS

2024	NSF Research Fellowship for Digital Transformation of Development
2021	Extraordinary Engineering and Technology Award, Lockheed Martin
2017	Tony Maxworthy Award for combined experimental and analytical excellence, USC
2017	Third Place, AIAA Region VI Student Paper Conference — Semi-truck Aerodynamic Drag Reduction

MENTORSHIP AND OUTREACH

Summer 2025	Graduate Mentor · UC Berkeley Transfer-to-Excellence (TTE) Research Program ► Designed and supervised independent research projects for two community-college students in robotics and AI: (i) modeling human behavior for human–robot interaction, and (ii) simulating dynamic fire environments for UAV disaster relief in Unity. ► Guided students through the full research arc — project design, data analysis, and poster preparation — culminating in presentations at the program’s research symposium.
Summer 2015	Mentor · Upward Bound, College of the Canyons

LEADERSHIP AND SERVICE

2026	Treasurer , Latino Association of Graduate Students in Engineering & Science (LAGSES), UC Berkeley
2025	Corporate Liaison , LAGSES, UC Berkeley
2016	Outreach Chair , Society of Hispanic Professional Engineers (SHPE), USC
2014	Founder & President , SHPE Chapter, College of the Canyons

PROFESSIONAL MEMBERSHIPS

Society of Hispanic Professional Engineers (SHPE) · Institute of Electrical and Electronics Engineers (IEEE) · Latino/a Association of Graduate Students in Engineering and Science (UC Berkeley) · Women in Robotics

SPECIALIZED SKILLS

Control & Optimization: Distributed optimization, game theory, mechanism design, classical control, trajectory optimization, safety-critical control.

Reinforcement Learning & ML: Multi-agent RL, deep RL (PPO), probabilistic modeling, Gaussian processes, supervised & unsupervised learning.

Robotics & Autonomous Systems: Multi-agent systems, UAV systems, human–robot interaction, ROS 1/2, Gazebo, AirSim, Isaac Sim, agent-based simulation.

Programming & Tools: Python, MATLAB, C++, Linux, Docker, Git.

CAD & Engineering Software: Siemens NX, SolidWorks, CREO, CATIA V5, STAR-CCM+, Thermal Desktop, SINDA, FEMAP.

Other Technical: Aerodynamics & CFD, experimental methods (wind-tunnel testing).